VINUNIVERSITY

Test duration: 1 hour and 30 minutes

Name: **Student ID:**

Midterm Practice Test 2 – Fall 2026

COMP3040/5030 Computer Vision Unofficial practice test, prepared by Lê Tăng Phú Quý

Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

About the exam: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

Problem	Content	Point (Undergraduates)	Type
1	Fundamentals of Computer Vision	20	short answer
2	Image Filtering	25	long answer
3	Feature Detection and Matching	25	long answer
4	Fitting and Image Transformations	25	long answer
5	System Design	05	long answer
Total		**100**	

Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

1.1. Distinguish image classification, object detection and semantic segmentation by the **output** of each task. (5 pts)

1.2. A 6.4 mm wide sensor has a 90° horizontal FOV. What is the focal length? If the focal length doubles, what is the new FOV? (5 pts)

1.3. What is **aliasing**? How do you avoid it when shrinking an image $\times 2$? (5 pts)

1.4. Explain the **aperture** trade-off of a pinhole camera and the role of a lens. (5 pts)

Problem 2. Image Filtering

2.1. (a) Show that the 2-D Gaussian kernel is **separable**. (b) For a 7×7 kernel, how many multiplications per pixel does direct 2-D filtering cost, and separable filtering? (c) Blurring with $\sigma = 6$ then $\sigma = 8$ equals a single blur with which σ? (12 pts)

2.2. Signal $f = [3, 1, 4, 1, 5, 9, 2]$, kernel $k = [1, 2, 3]$. Compute (a) the **correlation** and (b) the **convolution**, 'valid' mode. (c) Which one is commutative? Why does it matter when chaining filters? (13 pts)

Problem 3. Feature Detection and Matching

3.1. (a) Why does the DoG $L(k\sigma) - L(\sigma)$ approximate the scale-normalized Laplacian of Gaussian? (b) A bright disc of radius 6 px responds most strongly at which σ? (c) With $s = 3$ intervals per octave, what is k, and how many Gaussian and DoG images per octave? (10 pts)

3.2. A SIFT candidate has the spatial Hessian $H = \begin{bmatrix} 12 & 2 \\ 2 & 1 \end{bmatrix}$. With $r = 10$, keep or reject? Explain geometrically. (8 pts)

3.3. Descriptor $\mathbf{a}$ has three nearest neighbours in the other image at distances $0.21, 0.24, 0.65$. With a ratio test of 0.8, is its best match kept? Why is the ratio test better than an absolute threshold? (7 pts)

Problem 4. Fitting and Image Transformations

4.1. Three image points $(2, 0), (1, 1), (0, 2)$. (a) Write the line through them in normal form $\rho = x \cos\theta + y \sin\theta$ with $\theta \in [-90°, 90°)$. (b) What curve does each point produce in (θ, ρ) space? (c) Why use (ρ, θ) instead of (m, b)? (12 pts)

4.2. An affine map sends $(0, 0) \to (1, 2)$, $(1, 0) \to (3, 3)$, $(0, 1) \to (0, 4)$. (a) Find its 2×3 matrix. (b) The image of $(2, 2)$? (c) Is it a similarity (rotation + uniform scale + translation)? If so, find the angle and scale. (13 pts)

Problem 5. System Design

5. Design a phone **panorama mode**: the user sweeps the phone and takes 5 overlapping photos. Describe the automatic stitching algorithm and name at least two failure cases with mitigations. (5 pts)